

Precision Agriculture Enabling Farming Drones

We all have heard that drones can be used for agriculture. But how many of us are aware of the exact drones available for the purpose and their characteristics? And which drone would be the best for a particular application? Read on to find out what nobody may have told you so far

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Precision Agriculture is the key component for the third wave of modern agricultural revolution. The first agricultural revolution was of mechanised agriculture from 1990 to 1930, followed by the Green Revolution in 1960s with integration of new methods of genetic modification. Over the years, farming has evolved from merely growing crops to intensive use of farm equipment, and today the primary focus is integration of technology in farming. A recent study by Hexa Reports suggests precision agriculture is set to grow to \$43.4 billion by 2025.

Precision agriculture uses technological innovations to grow crops more efficiently. Its key component is the use of information technology and a wide array of technologies like GPS guidance, control systems, sensors, robotics, drones, autonomous vehicles, variable rate technology,

GPS based soil sampling, automated hardware, telematics, software, and of course precision machinery.

Drones assist farmers to constantly monitor crops and livestock, which could be quite cumbersome to check from the ground. According to Global Market Insights, the agricultural drone market is expected to grow from a \$1.2 billion in 2019 to \$4.8 billion in 2024. From scouting to security, drone use will become more ubiquitous on large- and small-scale farms in just a few years. The information gathered by drones on farms is often used for agronomic decisions.

In many areas, drone use has become an essential part of large-scale precision farming already. The data collected from drones help farmers plan their planting and treatments to achieve the best possible yields. Some reports indicate that using precision

farming systems can increase yields by as much as 5%, which is a sizeable increase in an industry with typically slim profit margins.

Sensors and digital imaging capabilities of drones give farmers a deeper insight in terms of irrigation problems, soil vibration, and pest cum fungal infections. Fig. 1 shows a drone helping in precision farming.

Why agricultural drone is important?

Here are some areas where drones can help in agriculture:

Plant monitoring. Drones with advanced special imaging called Normalised Difference



Fig. 1: Use of agricultural drone in a farm